

1. Title

CreatorIQ : Creator Analytics & Content Performance Dashboard

2. Objective

Build a creator analytics dashboard that helps content creators track social media performance, audience engagement, content growth, and monetization insights through a centralized platform.

The system supports analytics visualization, content performance tracking, audience insights, engagement monitoring, revenue analytics, and social media integration for multiple platforms.

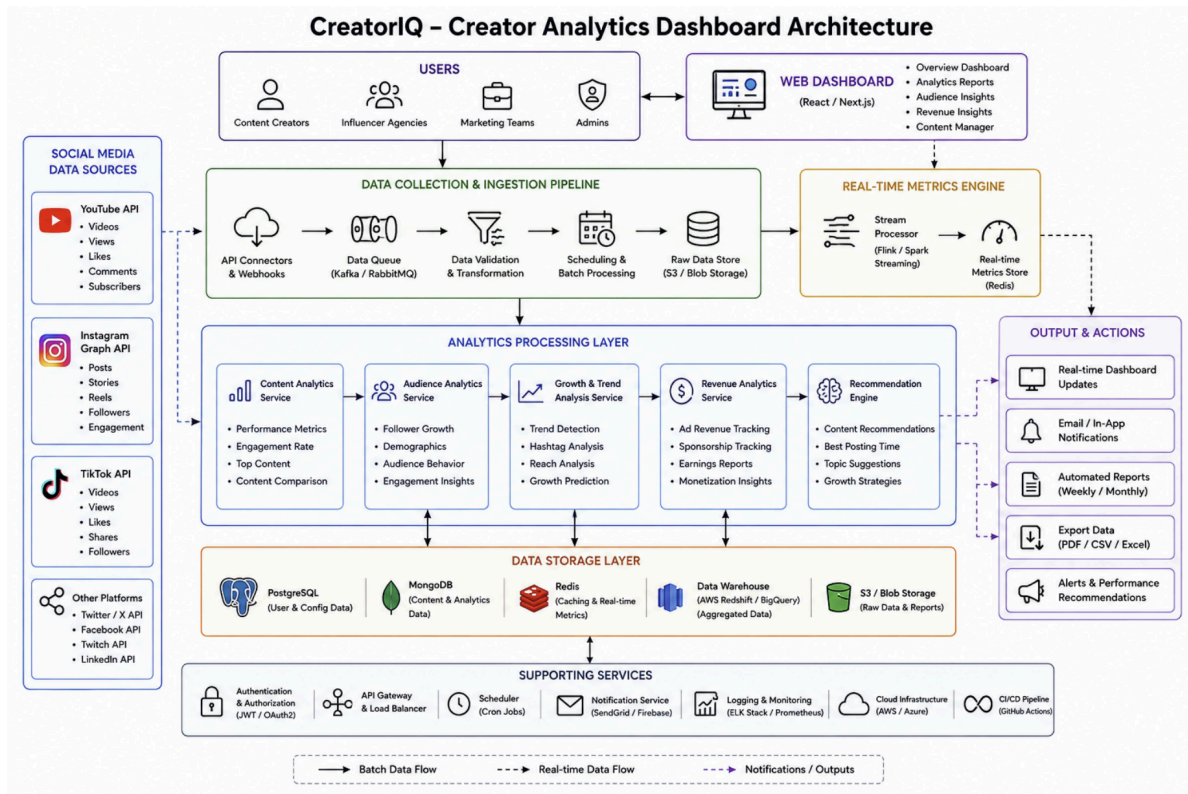
The platform is designed to help creators, influencers, agencies, and digital marketers make data-driven decisions and improve content strategy.

This solution can be used by YouTubers, Instagram creators, streamers, influencer agencies, marketing teams, and content management organizations.

Outcomes

- (i) Developed and deployed a creator analytics and content performance dashboard.
- (ii) Implemented social media analytics and engagement tracking systems.
- (iii) Built audience growth and demographic analysis workflows.
- (iv) Developed monetization and sponsorship tracking modules.
- (v) Implemented multi-platform social media integration workflows.
- (vi) Built analytics dashboards with interactive charts and KPI monitoring.
- (vii) Developed notification and exportable reporting systems.
- (viii) Built trend analysis and content comparison workflows.
- (ix) Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.

3. Architecture Components



4. Modules to be Implemented

1. User Management Module

- (i) User registration and login
- (ii) Creator profile management
- (iii) Agency profile management
- (iv) Role-based access control
- (v) Account settings

Roles

- Creator
- Agency
- Marketing Team
- Administrator

2. Content Analytics Module

- (i) Track content performance
- (ii) Engagement monitoring
- (iii) Content comparison dashboard
- (iv) Top-performing content reports
- (v) Reach analysis
- (vi) Performance trends

Metrics

- Views
- Likes
- Comments
- Shares
- Saves
- Watch Time
- Reach
- Engagement Rate

3. Audience Analytics Module

- (i) Follower growth tracking
- (ii) Audience demographics
- (iii) Audience engagement insights
- (iv) Reach analysis
- (v) Impressions tracking
- (vi) Audience behavior monitoring

Audience Data

- Age Distribution
- Gender Distribution
- Geographic Location
- Device Usage
- Active Hours

4. Growth & Trend Analysis Module

- (i) Growth monitoring
- (ii) Trend detection
- (iii) Hashtag analysis
- (iv) Reach prediction
- (v) Content growth tracking
- (vi) Audience growth forecasting

5. Revenue Analytics Module

- (i) Sponsorship tracking
- (ii) Ad revenue monitoring
- (iii) Earnings reports
- (iv) Revenue trends
- (v) Monetization analytics
- (vi) Financial insights

Revenue Sources

- Sponsorships
- Ad Revenue
- Affiliate Marketing
- Brand Collaborations
- Subscription Revenue

6. Social Media Integration Module

- (i) Connect social media accounts
- (ii) API-based data collection
- (iii) Multi-platform dashboard
- (iv) Scheduled synchronization
- (v) Content management integration

Supported Platforms

- YouTube
- Instagram
- TikTok
- Facebook
- X (Twitter)
- LinkedIn

7. Analytics Dashboard Module

- (i) Interactive charts
- (ii) KPI monitoring
- (iii) Performance comparison
- (iv) Real-time analytics
- (v) Trend visualization
- (vi) Growth dashboards

8. Notification & Reporting Module

- (i) Performance alerts
- (ii) Weekly analytics reports
- (iii) Engagement notifications
- (iv) Revenue alerts
- (v) Exportable reports
- (vi) Scheduled reporting

9. Reports & Export Module

- (i) Analytics reports
- (ii) Audience reports
- (iii) Revenue reports
- (iv) Growth reports
- (v) Platform comparison reports

(vi) PDF export

(vii) Excel export

10. Final Integration, Testing & Deployment

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

- (i) Define project objectives and analytics workflows.
- (ii) Design dashboard architecture and database schema.
- (iii) Create UI wireframes and dashboard workflow planning.
- (iv) Setup React frontend and FastAPI backend.
- (v) Implement authentication and role-based access system.
- (vi) Build creator dashboard UI.
- (vii) Develop analytics visualization components.

Outcomes

- (i) Understand analytics dashboard and creator management workflows.
- (ii) Learn system architecture and database design concepts.
- (iii) Build frontend and backend project initialization.
- (iv) Working authentication and analytics dashboard system.

Milestone 2: Week 3 & 4 — Content Analytics & Social Media Integration

Tasks

- (i) Implement engagement tracking workflows.
- (ii) Build audience analytics reports.
- (iii) Generate growth and trend charts.
- (iv) Connect social media APIs.
- (v) Fetch creator analytics data.
- (vi) Build multi-platform data management workflows.

Outcomes

- (i) Implement content and audience analytics systems.
- (ii) Build social media integration and API workflows.
- (iii) Understand analytics processing and engagement tracking concepts.
- (iv) Generate real-time creator performance insights.

Milestone 3: Week 5 & 6 — Revenue Analytics & Reporting

Tasks

- (i) Build earnings and sponsorship tracking system.
- (ii) Generate monetization reports and analytics.
- (iii) Implement analytics report generation workflows.
- (iv) Build notification and alert system.
- (v) Create exportable dashboard reports.

Outcomes

- (i) Build revenue and monetization analytics systems.
- (ii) Implement reporting and notification workflows.
- (iii) Understand analytics reporting and KPI monitoring concepts.
- (iv) Complete end-to-end creator analytics workflows.

Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

- (i) Perform application testing and workflow validation.
- (ii) Improve dashboard responsiveness and system optimization.
- (iii) Deploy platform using Docker and cloud environments.
- (iv) Prepare final project documentation and presentation.
- (v) Demonstrate the complete CreatorIQ platform.

Outcomes

- (i) Gain deployment and testing experience.
- (ii) Improve platform stability and usability.
- (iii) Complete live deployment and final demonstration.
- (iv) Prepare professional project documentation and presentation.

6. Evaluation Criteria

Milestone 1 (Week 2)

- (i) Project initialization and architecture setup completed.
- (ii) Authentication and dashboard management implemented.
- (iii) Analytics visualization components functional.
- (iv) System design and UI planning completed.

Milestone 2 (Week 4)

- (i) Engagement tracking and audience analytics implemented.
- (ii) Social media API integration functional.
- (iii) Growth and trend reports generated.
- (iv) Multi-platform workflows integrated.

Milestone 3 (Week 6)

- (i) Revenue tracking and monetization analytics implemented.
- (ii) Notification and reporting system functional.
- (iii) Exportable reports and dashboards working.
- (iv) KPI monitoring workflows integrated.

Milestone 4 (Week 8)

- (i) Fully deployed frontend and backend.
- (ii) Testing and validation completed.
- (iii) Documentation and presentation prepared.
- (iv) Successful end-to-end platform demonstration completed.

7. Tools & Tech Stack

Programming Language

Backend

- (i) Python
- (ii) FastAPI

Frontend

- (i) JavaScript
- (ii) React.js

Database

Primary Database

- (i) PostgreSQL

Secondary Database

- (i) MongoDB

Cache

(i) Redis

Analytics & APIs

(i) YouTube Data API

(ii) Instagram Graph API

(iii) TikTok API

(iv) Facebook Graph API

(v) LinkedIn API

Libraries & Frameworks

Backend

(i) FastAPI

(ii) SQLAlchemy

(iii) Pydantic

(iv) Alembic

(v) Celery

(vi) Redis

Frontend

(i) React.js

(ii) Tailwind CSS

(iii) Chart.js

(iv) Recharts

(v) Axios

(vi) React Router

Authentication

(i) JWT Authentication

(ii) OAuth2 Authentication

Cloud & DevOps

(i) Docker

(ii) Docker Compose

(iii) AWS / Azure

(iv) GitHub Actions

Dev & Deployment Tools

(i) VS Code

(ii) Git & GitHub

(iii) Postman

8. Performance Metrics

Analytics Performance

(i) Data processing speed

(ii) Dashboard loading time

(iii) Engagement tracking accuracy

Reporting Performance

(i) Report generation efficiency

(ii) API synchronization accuracy

(iii) Real-time analytics latency

System Performance

(i) API response time

(ii) Concurrent dashboard handling

(iii) Database query optimization

(iv) Cache efficiency

9. Example Quantitative Goals

Content Analytics

Generate accurate engagement and content performance insights.

Audience Analytics

Track audience growth and engagement behavior efficiently.

Revenue Monitoring

Generate reliable monetization and sponsorship analytics reports.

Platform Performance

Support multiple concurrent dashboard users with stable analytics performance.